

Exact Rational Certificates for the Reverse LCD–LP Comparison at $(20, 8)$

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Abstract

Kang and Xiong reported a floating-point screening observation at the binary parameter pair $(n, k) = (20, 8)$: their Gauss-phase relaxation is feasible at distance 7, whereas the mixed joint-weight-enumerator relaxation has endpoint 6. They explicitly excluded this reverse case from their exact endpoint claims. We certify the observation over the rationals. Specifically, for the systems in their Definitions 4.4 and 5.1, we prove

$$\mathcal{G}_2(20, 8, 7) \neq \emptyset, \quad \mathcal{M}_2(20, 8, 7) = \emptyset, \quad \mathcal{M}_2(20, 8, 6) \neq \emptyset.$$

The certificates are an integral point in an admissible Gauss branch, a sparse rational mixed point with 617 nonzero coordinates, and a dyadic Farkas vector whose 1,813 reduced-cost coordinates are strictly positive. Standalone exact-arithmetic checkers reconstruct every constraint, including all $1,771^2$ coefficients of the joint MacWilliams transform. These statements concern LP feasible regions only and imply neither existence nor nonexistence of a binary LCD code.

1 Context and statement

Linear programming bounds for weight distributions trace back to the MacWilliams transform. For LCD codes, Alahmadi, Deza, Dutour Sikirić, and Solé introduced a joint-weight-enumerator formulation involving a code and its dual [1]. Kang and Xiong subsequently defined a much smaller Gauss-phase relaxation and compared its endpoint with the mixed joint-enumerator benchmark [2]. Their computation found four exactly certified improvements in one direction. It also found the reverse comparison $(20, 8) : 7 \rightarrow 6$, but labelled that case a screening observation and did not include it among the exact endpoint claims.

The purpose of this note is solely to supply the omitted exact decision at that parameter pair. The use of finite, independently replayable certificates also occurs in adjacent coding and discrete-geometry calculations; see, for example, Zeng, Liu, Ratnavelu, and Ong [3]. That reference is cited only for this broad methodological comparison, not for any LCD constraint or claim proved here.

We now specify the systems so that the certificate claims have an unambiguous meaning. For $0 \leq j, w \leq 20$, put

$$K_j(w) = \sum_{\ell=0}^j (-1)^\ell \binom{w}{\ell} \binom{20-w}{j-\ell},$$

where an out-of-range binomial coefficient is zero. The common Hamming baseline at

dimension 8 consists of variables $A_w, B_w \in \mathbb{R}$ satisfying

$$A_0 = B_0 = 1, \quad A_w, B_w \geq 0, \quad (1)$$

$$A_w = 0 \quad (1 \leq w < d), \quad (2)$$

$$2^8 B_j = \sum_{w=0}^{20} A_w K_j(w) \quad (0 \leq j \leq 20), \quad (3)$$

$$A_w + B_w \leq \binom{20}{w} \quad (1 \leq w \leq 20). \quad (4)$$

The variables are pseudo-weight distributions; they need not be integral or realized by a code.

At $(20, 8)$, the Gauss relaxation is the union over the eight admissible branches

$$(O, 0, 0), (O, 2, 0), (O, 4, 0), (O, 6, 0), (O, 0, 1), (O, 4, 1), (E, 0, 0), (E, 4, 0).$$

For a branch (τ, β, η) , one adds $A_{20} = \eta$ and

$$\begin{aligned} \sum_{w \equiv 0 \pmod{4}} A_w &= \frac{2^8 + \Delta_\tau + 2R_\beta}{4}, & \sum_{w \equiv 1 \pmod{4}} A_w &= \frac{2^8 - \Delta_\tau + 2I_\beta}{4}, \\ \sum_{w \equiv 2 \pmod{4}} A_w &= \frac{2^8 + \Delta_\tau - 2R_\beta}{4}. \end{aligned} \quad (5)$$

Here $\Delta_O = 0$, $\Delta_E = 2^8$, and, for the even phases used above,

$$(R_0, I_0) = (16, 0), \quad (R_2, I_2) = (0, 16), \quad (R_4, I_4) = (-16, 0), \quad (R_6, I_6) = (0, -16).$$

The resulting union is denoted by $\mathcal{G}_2(20, 8, d)$.

For the mixed relaxation let

$$I_{20} = \{(a, b, c, e) \in \mathbb{Z}_{\geq 0}^4 : a + b + c + e = 20\}, \quad |I_{20}| = 1771,$$

and append nonnegative variables $M_{a,b,c,e}$ to the baseline. In addition to (1)–(4), impose

$$M_{a,b,c,e} = 0 \quad (e \text{ odd}), \quad (6)$$

$$M_{a,b,c,e} = 0 \quad (1 \leq c + e < d), \quad (7)$$

$$M_{a,0,0,e} = 0 \quad (e > 0), \quad (8)$$

$$M_{20-w,w,0,0} = B_w, \quad M_{20-w,0,w,0} = A_w \quad (0 \leq w \leq 20), \quad (9)$$

$$\sum_{c+e=w} M_{a,b,c,e} = 2^{12} A_w \quad (0 \leq w \leq 20), \quad (10)$$

$$\sum_{b+e=w} M_{a,b,c,e} = 2^8 B_w \quad (0 \leq w \leq 20), \quad (11)$$

$$\sum_{b+c=j} M_{a,b,c,e} = \binom{20}{j} \quad (0 \leq j \leq 20), \quad (12)$$

where each sum ranges over I_{20} . Finally, with

$$J(x) = \sum_{(a,b,c,e) \in I_{20}} M_{a,b,c,e} x_0^a x_1^b x_2^c x_3^e, \quad H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

require the coefficientwise polynomial identity

$$J(x) = 2^{-20} J(H_4 x). \quad (13)$$

This feasible region is $\mathcal{M}_2(20, 8, d)$. Increasing d only adds equations to the zero family (7), so these mixed regions are nested. We call the largest feasible value of d the mixed endpoint.

Theorem 1. *For the real LP regions just defined,*

$$\mathcal{G}_2(20, 8, 7) \neq \emptyset, \quad \mathcal{M}_2(20, 8, 7) = \emptyset, \quad \mathcal{M}_2(20, 8, 6) \neq \emptyset.$$

The Gauss point belongs to the $(O, 0, 0)$ branch. Consequently, the endpoint of the mixed relaxation at $(20, 8)$ is exactly 6. No conclusion about the existence of a binary LCD code follows from these LP statements.

2 The integral Gauss witness

Let $A(z) = \sum_w A_w z^w$ and $B(z) = \sum_w B_w z^w$. The following table gives all nonzero coordinates; every omitted coordinate is zero.

w	0	7	8	9	10	11	12	13	14
A_w	1	36	53	28	36	28	18	36	20

w	0	2	5	6	7	8	9	10	11	12	13	14	15	16
B_w	1	1	84	203	260	440	680	727	680	524	260	125	84	27

All entries are nonnegative, $A_1 = \dots = A_6 = 0$, and direct integer expansion gives

$$(1+z)^{20} A \left(\frac{1-z}{1+z} \right) = 2^8 B(z), \quad (14)$$

which is equivalent to all 21 equations in (3). The displayed entries also satisfy each coefficientwise packing inequality $A_w + B_w \leq \binom{20}{w}$.

For the $(O, 0, 0)$ branch, $A_{20} = 0$ and the three residue sums are

$$1 + 53 + 18 = 72, \quad 28 + 36 = 64, \quad 36 + 20 = 56.$$

These are exactly the right sides of (5), namely $(256 + 32)/4$, $256/4$, and $(256 - 32)/4$. The branch is admissible: its dual type is O and its dual phase condition is $(20 - 0) \bmod 8 = 4 \in B_O(12)$. Thus the table is an integral feasible point and proves the first assertion of Theorem 1.

3 The mixed certificates

We use the variable order

$$A_0, \dots, A_{20}, B_0, \dots, B_{20}, \quad M_s \ (s \in I_{20} \text{ in lexicographic order}). \quad (15)$$

For $s, t \in I_{20}$ define the integer

$$T_{t,s} = [x^t] \prod_{i=0}^3 \left(\sum_{j=0}^3 (H_4)_{ij} x_j \right)^{s_i}. \quad (16)$$

Then (13) is represented by the 1771 integer rows

$$2^{20} M_t - \sum_{s \in I_{20}} T_{t,s} M_s = 0 \quad (t \in I_{20}). \quad (17)$$

3.1 A rational point at distance six

The ancillary file `evidence/mixed_d6_rational_point.json` is a complete sparse rational vector in the order (15). Each stored value is a pair of decimal integers $[p, q]$ denoting p/q with $q > 0$, and every unstored coordinate is exactly zero. It contains 617 nonzero coordinates among the 1813 variables, all strictly positive. The least common denominator has 333 bits.

For completeness, we describe the finite verification rather than appealing to an LP solver. The checker enumerates I_{20} , constructs the Krawtchouk rows from their defining binomial sum, and constructs each column of T by repeated integer multiplication by the corresponding row-linear form of H_4 . It then clears the common denominator once and evaluates every row using integers. For $d = 6$ it obtains 3099 equality rows, including every one of the $1771^2 = 3,136,441$ entries of T , together with 20 packing rows. Every equality has zero residual, every packing slack is nonnegative, and every coordinate is nonnegative. This exact substitution proves $\mathcal{M}_2(20, 8, 6) \neq \emptyset$.

The certificate is large but finite and explicit; the checker does not rely on floating-point tolerances or solver status. Its standalone implementation and machine-readable result are `evidence/verify_primal_certificates_independent.py` and `evidence/primal_certificates_verification.json`.

3.2 A Farkas certificate at distance seven

Write the 3205 equality rows of the $d = 7$ mixed system as

$$Ex = b, \quad x \geq 0, \quad (18)$$

using (15); the 20 packing inequalities will not be needed. The file `evidence/mixed_d7_farkas.json` gives a rational vector y with 1703 nonzero equality multipliers. With

$$D = 2^{86} = 77371252455336267181195264,$$

all its entries have the declared dyadic representation, and exact coefficientwise multiplication gives

$$s := E^T y > 0, \quad \min_j (Ds_j) = 2^{52}, \quad (19)$$

for all 1813 coordinates. The same exact calculation gives

$$y^T b = -\frac{73714918700799625629}{73786976294838206464} < 0. \quad (20)$$

If a vector x satisfied (18), then

$$0 \leq s^T x = (E^T y)^T x = y^T (Ex) = y^T b < 0,$$

a contradiction. Hence even the system obtained by deleting the packing inequalities is infeasible, and therefore $\mathcal{M}_2(20, 8, 7) = \emptyset$.

The standalone checker `evidence/verify_d7_farkas_independent.py` reconstructs E from the displayed definitions, generates all 3,136,441 joint-transform coefficients, matches the named row indices, recomputes every coordinate of (19), and evaluates (20) in integer arithmetic. Its machine-readable report is `evidence/mixed_d7_farkas_verification.json`. This completes the proof of Theorem 1.

4 Scope and reproducibility

The exact delta from the inspected Kang–Xiong preprint is restricted to its single reverse screening case: the three certificates above replace a floating-point observation by rational feasibility and infeasibility proofs. They do not certify a broader comparison between the two relaxations. Moreover, LP pseudo-distributions may fail to arise from codes, so neither positive certificate constructs an LCD code and the negative certificate rules out only the specified mixed LP at $d = 7$.

The ancillary JSON files use decimal integers only. From the project root, the two decisive replays are run as

```
python evidence/verify_primal_certificates_independent.py
python evidence/verify_d7_farkas_independent.py
```

Both programs use exact integer/rational arithmetic for the acceptance checks. Floating-point optimization was used only to locate candidate certificates and is not part of the proof.

OpenAI Codex assisted with computational exploration, script integration, exact-certificate preparation and checking, and manuscript drafting. The author reviewed the definitions, certificates, verification outputs, and final mathematical claims. The explicit rational data and exact finite replays, not the output or confidence of an AI system, constitute the evidence.

References

- [1] Adel Alahmadi, Michel Deza, Mathieu Dutour Sikirić, and Patrick Solé. The joint weight enumerator of an LCD code and its dual. *Discrete Applied Mathematics*, 257:12–18, 2019. <https://doi.org/10.1016/j.dam.2018.10.032>.
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- [3] Zijian Zeng, Houde Liu, Kuru Ratnavelu, and Seng Huat Ong. Exact finite certificates for the four-dimensional ternary borsuk problem and five-dimensional ternary kissing codes. *Preprints*, August 2026. <https://doi.org/10.20944/preprints202608.1658.v1>.